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PALEOMAGNETIC DATA OF MIDDLE AND UPPER TRIASSIC AGE FROM THE DOLOMITES (EASTERN ALPS, ITALY)

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SUMMARY

A paleomagnetic survey of Middle and Upper Triassic porphyrites, dykes, and tuffites from the Dolomites (Eastern Alps, Italy) has yielded an average direction (based on 57 samples from 12 collecting localities) with declination 329°E of true north and inclination +25° after magnetic cleaning and correction for geological dip. The declination deviates from that of "stable" Northern Europe by about 50°, but the inclination agrees well with the expected value. This indicates that the Dolomites area has rotated 50° anticlockwise relative to northern Europe since the Triassic but there is no evidence of relative changes in latitude. A few new results from the Predazzo complex, a Triassic igneous group, and from the Valgardena Red Beds of Middle and Upper Permian age, are also described.

INTRODUCTION

Paleomagnetic observations in the Alps and other parts of the Tethys are inconsistent with results from "stable" regions of Northern Europe and Northern Asia (Clegg et al., 1957; Van der Lingen, 1960; Dietzel, 1960; Van Hilten, 1962; Schwarz, 1962, 1963; De Boer, 1963, 1965; Guicherit, 1964; Van Hilten and Zijderveld, 1966; Van Dongen, 1967). These deviations imply, in a general fashion, that large scale relative motions, either rotational or translational, have occurred between Tethys and its northern borderland. But much more data than is presently available is needed before these apparent motions can be evaluated quantitatively. This paper describes new observations (chiefly from Triassic igneous rocks) from the area of the Dolomites which is within the Tethys domain, and is intended as a contribution to this necessary body of knowledge. The Dolomites consist of little disturbed Permian and Mesozoic formations resting on a crystalline basement complex. There is no evidence of regional metamorphism and nappe tectonics, and if large displacements have occurred they have almost certainly affected the region as a whole. It was for these reasons that this sampling region was chosen.

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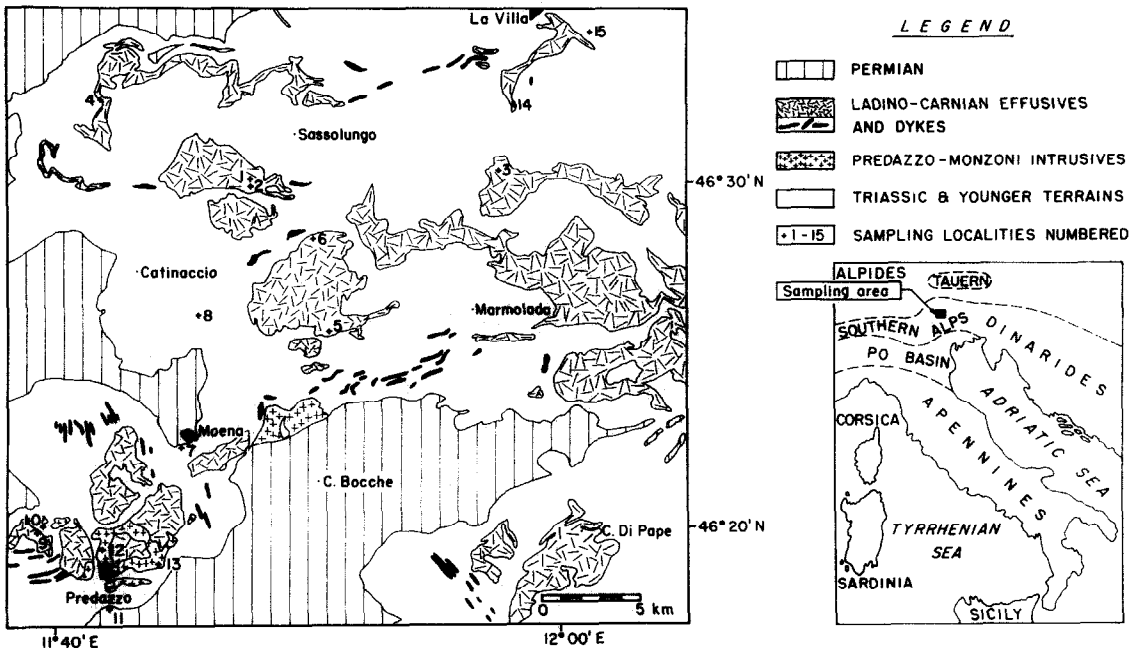


Fig.1. Location map. The localities are numbered as follows:
Triassic porphyrites: *1* = Upper Duron; *2* = Lower Duron; *3* = Campolongo; *4* = Alpe di Siusi; *5* = Ciampie; *6* = Campitello; *9* Monte Agnello I and II.
Triassic dykes: *7* = Moena; *8* = Ciampedie I and II; *10* = La Porta; *11* = Malgola.
Predazzo intrusive complex: *12* = Predazzo; *13* = Zaluna.
Triassic tuffites: *14* = Corvara; *15* = S. Cassiano.

GEOLOGY AND SAMPLING

Four Triassic and one Permian formations have been sampled. The sampling localities are plotted in Fig.1 and those of the Permian "Valgardena sandstones" are described below.[1]

Triassic porphyrites

These lavas, recently described by Simboli (1966), are stratigraphically well dated, being associated with fossiliferous tuffites (Leonardi, 1968). Their ages range from Upper Ladinian to Lower Carnian. They occur as submarine, flat-lying flows, up to 70 m thick. Pillowlavas are common. Most of the results are from this unit. Chatterjee (1965) has also reported results from this.

Triassic dykes

These dykes are associated with the porphyrites and are of similar composition. They are generally considered to be of the same age. They have been sampled only when cutting sedimentary beds of low dip, so that their original attitude can be reliably reconstructed.

La Valle Formation

This is a sedimentary formation consisting of arenites, shales, and tuffites, of Middle-Upper Ladinian age. Two localities of dark silty tuffites (henceforth referred to as Triassic tuffites) have yielded suitable samples.

Predazzo-Monzoni Complex

This complex locally intrudes and metamorphoses Anisian and Lower Ladinian limestones, so that its maximum age is Lower Ladinian. The complex was often thought to be Tertiary, but recent K-Ar age determinations have yielded an age of 230 m.y. (Borsi and Ferrara, 1967), which is Triassic. Several rock types were sampled but only two basic dykes (henceforth referred to as Predazzo dykes) are sufficiently strongly

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[1] Soraga: Reddish siltstones interbedded with sandstones in a stream at the village of Soraga. Locality about 300 m W of the village. Sheet 11 (I.G.M., Istituto Geografico Militare) 11°36'35"E, 46°22'50"N.
Passo M. Croce: Purple siltstones and clay stones on the Strada Statale 52, 300 m, SE of Passo di Monte Croce. Beds dip 49° at 134°. Sheet 12 (I.G.M.) 12°22'15"E, 46°28'50"N.
S. Martino: Pink micaceous sandstones and siltstones in a stream 500 m of Piccolino, in Val Badia. Beds dip 18° at 075°. Sheet 4A (I.G.M.) 11°55'15"E, 46°37'15"N.
Trento: Dark red fine-grained sandstones and siltstones with clay modules. Locality near Trento at the start of the road to Gardolo. Subvertical beds with mean strike direction 263°. Sheet 21 (I.G.M.) 11°09'35"E, 46°06'00"N.

magnetized for measurement. These dykes lack tectonic control. However all authors (see for example Leonardi, 1968) agree that little tilting has taken place, so that it is reasonable to refer their directions to the present horizontal plane.

Valgardena Red Beds

This Middle to Upper Permian formation was sampled at 17 localities in the Eastern Alps. Only two yielded satisfactory data.

RESULT FROM TRIASSIC SAMPLES

The susceptibilities of samples were measured on a commercial bridge. The values for porphyrites and dykes range from 1.1 to 7.6 · 10^{-3} c.g.s./cm^3, and show a two-peak distribution, the peaks having mean values of about 4.5 and 2.0. The tuffites have susceptibilities of between 0.7 and 1.5 with a mean of 1.2 · 10^{-3} c.g.s./cm^3.

Directions and intensities of remanent magnetization were measured with both astatic and spinner magnetometers. Alternating field (A.F.) demagnetization curves for one specimen from each site are given in Fig.2. Except for results from two sites, detailed in the legend of Fig.2A, curves of igneous rocks fall sharply, and are contained within a well-defined envelope. The magnetization is for the most part not of a very stable type, and there are large components with low coercivities.

The natural remanent magnetization (N.R.M.) intensities of the Triassic porphyrites average 4 · 10^{-3} c.g.s./cm^3 with several samples as strong as 13 · 10^{-3}. The intensities fall to less than 10% after demagnetization in 400 Oe alternating field. In the two deviating curves of Fig.2A the intensities are much more stable and decrease only to 30% of the initial value. Triassic dykes and tuffites (Fig.2B, 2C) have much lower initial intensities (means of 0.7 and 0.5 · 10^{-3} e.m.u. respectively) but the relative decay under alternating field demagnetization is the same as for the lavas.

In low demagnetizing fields substantial directional changes occur, but at field of 200–300 Oe the directions stabilize, indicating that a small high coercivity component is present. Beyond 500 Oe the directions begin to scatter. Typical direction migration for one specimen is shown in Fig.3. The effect on directions of partial demagnetization at one locality is given in Fig.4. The initial wide scatter (k = 5) is reduced after cleaning (k = 28) in 300 Oe, but the scatter is still substantial, and does not decrease in higher fields. This scatter is partly orientation error, but it is attributed also to "noise" caused by the presence of large components of low coercivity material which developed randomly directed magnetization during processing. Nevertheless, a component, systematically divergent from the present field, is clearly discernible. The directions of all Triassic samples before and after cleaning are plotted in Fig.5. The original scatter is reduced after cleaning in 300 Oe and the directions group in the NW direction. There is also a small group in the NE quadrant. These samples yield the unusual decay curve of Fig.2A.

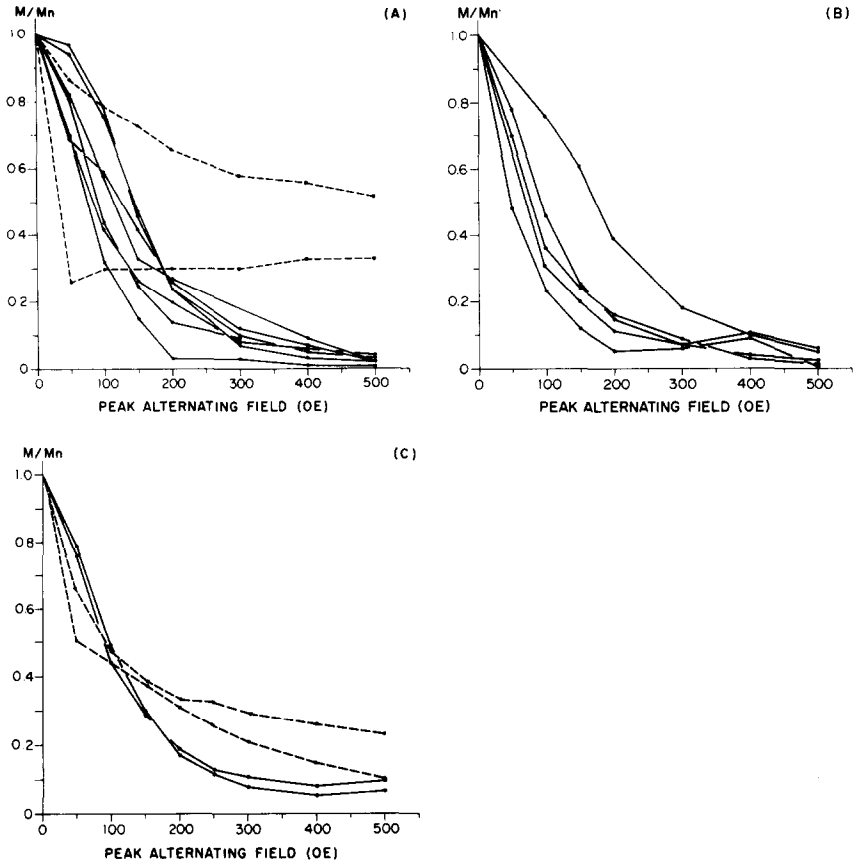


Fig.2. Alternating field demagnetization curves. A. Triassic porphyrites; dashed lines are from Monte Agnello I and II. B. Triassic dykes. C. Triassic tuffites (full lines) and Predazzo dykes (dashed lines). Relative intensity is plotted on the vertical axis.

The results have been analyzed statistically using the methods of Fisher (1953) and the computer program of Grieve and Irving (1968). The site means are tabulated in Table I and grand averages in Table II. Certain data have not been included in the analysis: three samples from Campitello for their unstable magnetization; the samples from Moena because the tectonic correction is unreliable being close to a fault; the samples from site Monte Agnello (I) because their directions are close to the present field's direction and have unusual decay characteristics (Fig.2A). The directions in the NW quadrant are from 12 sites in four formations with different tectonic attitudes. The agreement of directions after tectonic correction constitutes a successful fold test. Both tectonic correction and magnetic cleaning caused a systematic migration of directions away from

TABLE I

Locality means results[1]

Localities	N	S	N.R.M.				After cleaning			
			M_n[1]	D, I	k	R	M	D, I	k	R
Triassic porphyrites										
Upper Duron	5	16	7.32	338 +16	14	4.72	0.49	335 +08	24	4.83
Lower Duron	5	11	4.86	337 +18	122	4.97	0.32	340 +11	71	4.94
Campolongo	5	21	3.70	276 +17	5	4.24	0.16	316 +12	28	4.85
Alpe die Siusi	6	21	4.48	001 +27	167	5.97	0.33	344 +19	67	5.92
Ciampie	4	12(9)	1.49	001 +36	125	3.98	0.07	319 +31	31	3.90
Campitello	5(2)	15(7)	1.11	289 +30	4	3.94	0.35	317 +15	25	1.96
M. Agnello I	5	15	0.81	002 +36	5	4.11	–	– –	–	–
M. Agnello II	5	13	3.65	038 –49	2	4.30	0.27	40 +10	15	4.70
Triassic dykes										
Malgola	6	14	0.45	289 +57	4	4.63	0.04	317 +45	16	5.68
Moena	2	2	0.79	275 +30	10	1.90	–	–	–	–
Ciampedie I	5	16	1.11	320 +60	4	3.94	0.06	312 +29	15	4.74
Ciampedie II	5	15	0.93	011 +60	3	3.55	0.04	333 +38	20	4.80
La Porta	5	10	0.36	022 +37	86	4.95	0.06	353 +36	300	4.99
Triassic tuffites										
Corvara	4	11	0.53	353 +45	7	3.58	0.06	318 +29	4	3.26
S. Cassiano	5	16	0.45	010 +33	8	4.53	0.05	344 +24	9	4.55
Predazzo dykes										
Avisio	4	11	1.30	– –	–	2.99	0.31	348 +31	8	3.60
Zaluna	5	13	0.86	346 +47	5	4.26	0.12	319 +53	26	4.84
Valgardena Red Beds										
Soraga	2	3	0.002	164 +21	16	1.94	0.001	156 –12	87	1.99
Passo M. Croce	3	17	0.019	141 –15	22	4.82	0.014	136 –19	28	4.86
S. Martino	1	2	0.004	118 +09	–	1	0.001	133 –07	–	–
Trento	2	6	0.003	167 –22	58	1.98	0.003	165 –11	225	1.99

[1] N = is the number of hand samples, S = the number of specimens (where the numbers differ after cleaning they are put in brackets), M = is the mean (geometrical) intensity of magnetizatio in units 10^{-3} c.g.s./cm^3, D and I are the declination and inclination, k = Fisher's precision factor, R is the vector sum of sample mean directions. All directions are given after tectonic correction, except for the Predazzo dykes where they are given with respect to horizontal. N.R.M. is the natural remanent magnetization. Cleaning has been carried out at 300 Oe peak AF for the igneous rocks and at 650°C for the Valgardena sediments.

TABLE II

Summary of results before and after tectonic correction (TC)[1]

Formation	Sites	N.R.M.		α	After cleaning		α	Pole
		before TC D, I	after TC D, I		before TC D, I	after TC D, I		
Triassic porphyrites	6	332 +41	332 +30	26	329 +24	329 +16	12	123W 44N
Triassic dykes	4	353 +59	352 +58	34	339 +50	329 +50	18	112W 55N
Triassic tuffites	2	349 +41	002 +39	–	330 +29	331 +27	–	122W 50N
Triassic average	12	338 +42	335 +42	17	329 +29	329 +25	9	120W 48N (10, 5)
Predazzo dykes	2	000 +52	–	–	336 +43	–	–	– –
Valgardena Red Beds	4	138 −09	147 −02	36	141 −28	148 −13	18	– –

[1] α is Fisher s 95% cone of confidence in degrees. The underlined figures are considered to be the most useful values summarizing the information given in this paper.

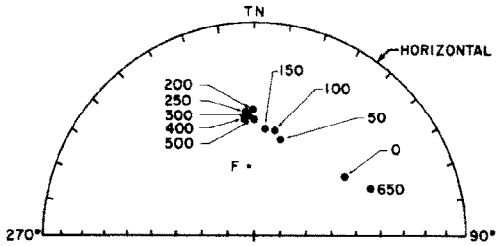


Fig.3. Directions changes in one specimen from a Predazzo dyke during alternating field demagnetization. The number at each point gives the peak demagnetizing field. *F* is the present field direction.

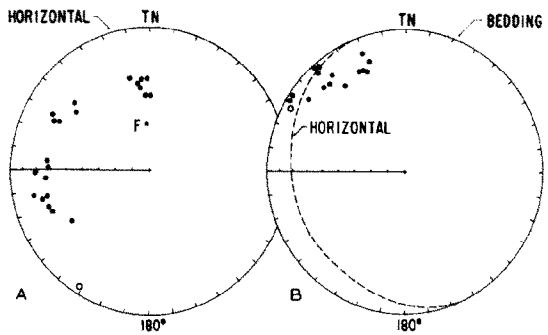


Fig.4. Direction changes in specimens from one locality of Triassic porphyrites (Campolongo). A. N.R.M. before tectonic correction. B. after cleaning in 300 Oe A.F. and tectonic correction. *F* is the present field.

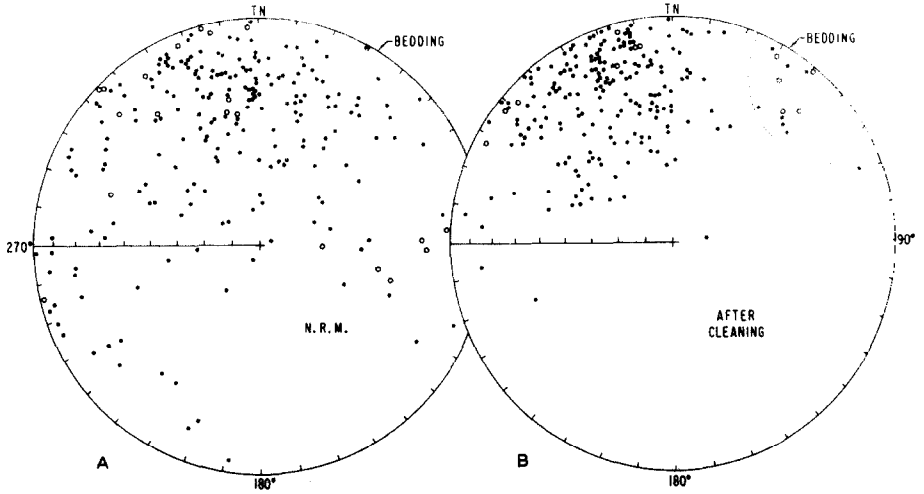


Fig.5. Directions of all Triassic specimens.
A. N.R.M., after tectonic corrections (220 specimens). B. After cleaning in 300 Oe field, after tectonic correction (214 specimens); discarded samples omitted as explained in text. The dashed line contains the unusual NE directions from Monte Agnello II. Open circles (solid dots): negative (positive) inclinations.

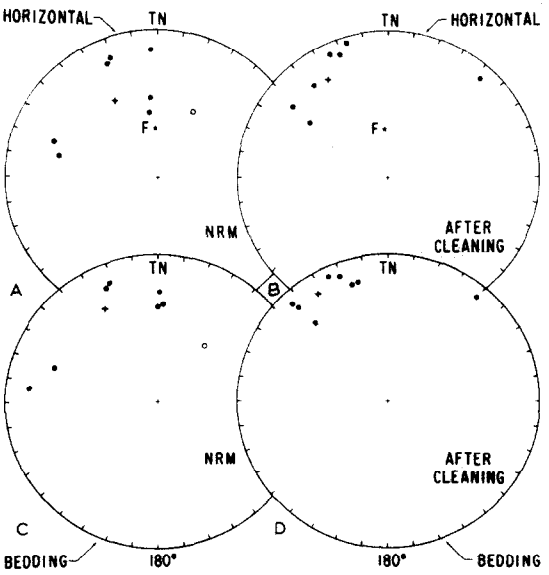


Fig.6. Locality mean directions in Triassic porphyrites. A, B. Before tectonic correction. C, D. After tectonic correction. F is the present field. The cross denotes the overall mean direction.

the present field; each formational mean direction decreases in declination and in inclination (Table II).

The low inclination of the Triassic porphyrites is considered the most reliable estimate because the porphyrites constitute the largest and most uniform group of samples, they have the highest precision, and their tectonic control is good. The effect of cleaning and tectonic correction on the porphyrites sites directions is shown in Fig.6.

Directions from Triassic dykes (Fig.7) are more scattered than those from the lavas. The directions change more after dip correction than after cleaning, and the dispersion is higher between samples from the same locality than between different localities. These facts indicate that the dyke results are less reliable. The Triassic tuffites (Fig.7) are the smallest group and have rather low precision. However, the mean direction lies very close to the Triassic grand average. The high within-site scatter may be due to the sedimentary nature of the material.

As for the Predazzo dykes (Fig.7) the directions of N.R.M. are highly scattered about the present field and after cleaning moves slightly toward the other Triassic directions. The scatter remains high ($k = 14$). One site (Avisio) has random directions.

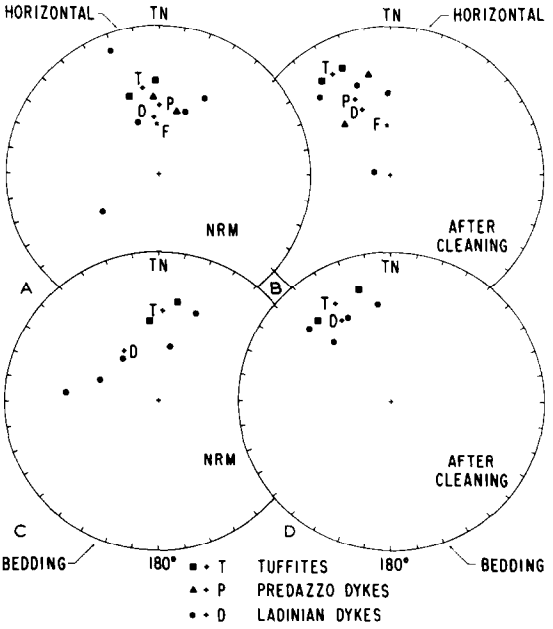


Fig.7. Locality mean directions of Triassic dykes (full circles), Triassic tuffites (squares), and Predazzo dykes (triangles). The crosses denote the respective means. F is the present field. A, B. Before tectonic correction. C, D. After tectonic correction. The Predazzo dykes are excluded from C and D because they lack accurate tectonic control.

The locality Monte Agnello II has a deviating direction (D = 40°, I = +10°) and all N.R.M. directions have negative inclinations which become positive upon demagnetization. The demagnetization curve (Fig.2) is also unusual. The meaning of this result is not understood.

RESULTS FROM THE VALGARDENA SANDSTONE

A collection from Permian red beds (Valgardena Formation) from the Eastern Alps has given scattered N.R.M. results (Fig.8). Thermal demagnetization of pilot samples from all localities at 500°, 550°, 600°, and

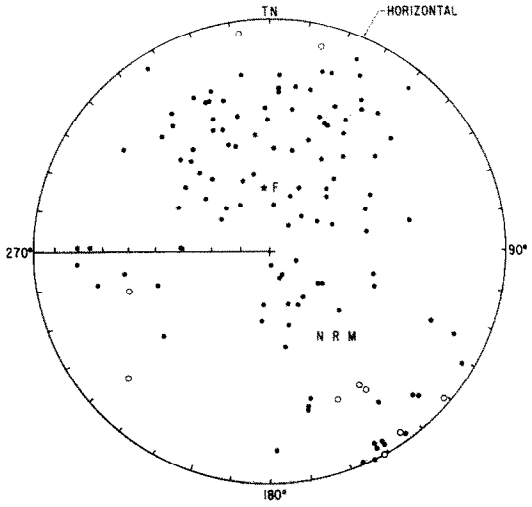


Fig.8. N.R.M. directions of magnetization of all specimens from the Valgardena sandstones before tectonic correction (120 specimens).

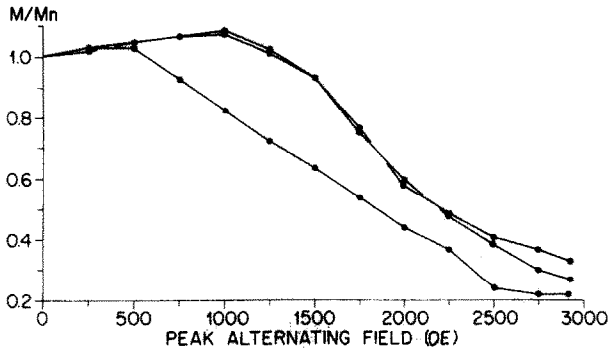


Fig.9. Alternating field demagnetization curves of three specimens from the Valgardena sandstones. Relative intensity is plotted on the vertical axis.

650°C showed nearly all of them to be unstable. However at four localities, although the directions showed large direction changes, these changes were systematically away from the present field direction, and in the temperature range 500–650°C the directions remained constant within the accuracy of the determinations (5° at these low intensities). There is a well-defined stable component present, directed toward the SE with low inclination. The rocks at the four localities were siltstones or very fine sandstones, whereas the rest of the collection was of coarser grain size. In Fig.9 the decay curves obtained by AF demagnetization of these stable specimens

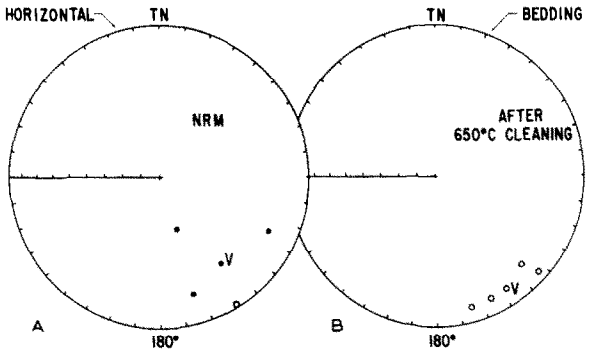


Fig.10. Locality mean directions of Valgardena sandstones. A. N.R.M. before tectonic correction. B. After thermal cleaning at 650°C and tectonic correction. V = overall mean direction.

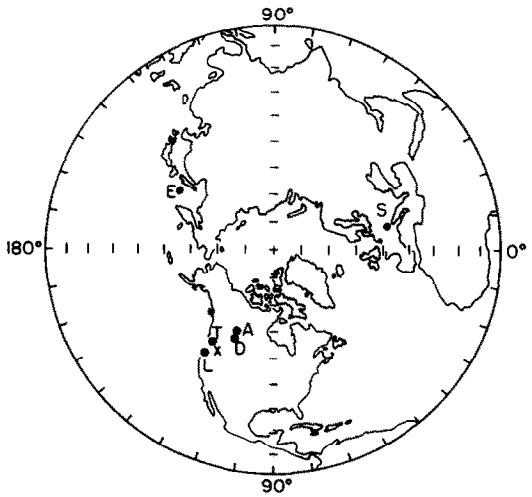


Fig.11. Triassic pole positions. *E* = stable Europe, (Irving, 1967). *A* = Triassic Alpine pole after Van Hilten (1964). Poles from Triassic porphyrites = *L*; Triassic dykes = *D*; Triassic tuffites = *T*, the cross marking the mean pole from these three formations. *S* is the sampling area.

are shown. The initial rise indicates the presence of a component in the direction along the present field. Strong high coercivity components are present. Directions are unchanged up to 2,000 Oe and change only by a few degrees at higher fields. All specimens from the four localities were thermally cleaned at 650°C. The drop of intensity at 650°C roughly corresponds to that at 1,500 Oe in A.F. demagnetization. The N.R.M. intensities of Valgardena sediment average about $5 \cdot 10^{-6}$ c.g.s./cm^3. The site directions are given in Table I and plotted in Fig.10. The bedding dips vary from 18°to 80° so the fact that the agreement after tectonic corrections is good (Fig.11) constitutes a successful fold test.

DISCUSSION

The Triassic formation means show good agreement between themselves (Table II), and it is reasonable to calculate an overall average. The results from the Predazzo dykes which lack tectonic control, and the aberrant results from Monte Agnello II, are excluded from this average, which is the mean of the porphyrites, dykes, and tuffites, giving each site unit weight.

A comparison with other Middle and Upper Triassic data from the Alps is made in Table III. Certain of the results (given in brackets) are regarded as unreliable by the authors concerned and are not considered for discussion. The results are not consistent amongst themselves. There are discrepancies in declination. Declinations in the NE quadrant were observed by De Boer (1963) in 3 Norian samples from the Vicentinian Alps, by Guicherit (1964) in 28 samples from the Carnian Alps, by Chatterjee (1965) in 4 samples from the Dolomites. Some NE declinations have also been found in the present study (Monte Agnello II). However most

TABLE III

Comparison of Middle and Upper Triassic paleomagnetic data from the Alps

Regio	Area	Reference	Age[1]	N	D, I
Carnian Alps	Tarvisio	Guicherit (1964)	L–C	28	028 +43
	Dobratsch	Van Lookeren (1960)[2]	L	8	(018 +50)
Dolomites	Fontanazzo	Guicherit (1964)	L	4	026 +29
	Dolomites	Chatterjee (1965)	L–C	4	026 +24
	Dolomites	this paper	L–C	57	329 +25
	Predazzo	this paper	L?	9	(336 +43)
Vicentinian Alps	Schio	De Boer (1963)	A	9	331 +49
	Schio	De Boer (1963)	L	7	330 +49
	Schio	De Boer (1963)	L	6	(343 +45)
	Schio	De Boer (1963)	NR	3	015 +47

[1] The ages are as follows: A = Anisian, L = Ladinian, C = Carnian, NR = Norian. N is the number of samples on which the result is based. Mean directions D, I are given after magnetic cleaning and tectonic correction. Data entered in brackets was considered unreliable for various reasons by the original authors.
[2] Cited in Guicherit (1964).

of the declinations observed by De Boer (16 samples) and in this study (57 samples) lie in the NW quadrant. Furthermore the inclinations are not in good agreement. Inclinations as high as +50° have been found in previous research (with the exception of Chatterjee's 4 samples) which do not correspond with the Triassic average inclination of +25° (error 9°) found from the Dolomites in this study. The reasons for these internal inconsistencies in results from the Triassic of the Alps will not become clear until very much more data is obtained.

The results reported herein are now compared with results from stable northern Europe. The mean pole (Fig.11) for Triassic formations (240°E, 48°N) is significantly different from the Triassic pole from northern Europe (146°E, 50°N, Irving, 1967). The difference can be entirely explained by an anticlockwise rotation of the area of the Dolomites of about 50° relative to "stable" northern Europe. That is, the significant difference is in declination, whereas the inclination is not significantly different from that of northern Europe. On the basis of this evidence there is no need to assume a large latitudinal displacement of the Dolomites relative to northern Europe since the Triassic, as De Boer (1965) suggests.

Since the preparation of this paper a new result (161, –21) based on 5 Triassic samples from one site in the Lombardy Alps (De Jong and Zijderveld, 1969) has appeared and is in substancial agreement with this conclusion. Similarly, the data presented here from the Valgardena Sandstone, although only based on 4 localities (10 samples), is inconsistent in declination but *not* in inclination with that from northern Europe.

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